

Jaden Ong

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EDUCATION

San Diego State University — *B.S. in Computer Science, Minor in Mathematics* *Expected May 2027*
GPA: 3.96 — *Data Structures & Algorithms, Cloud Computing, Database & Web Programming, Machine Learning, Operating Systems, UNIX System Administration*

SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript/TypeScript

Frameworks: Next.js, FastAPI, Spring Boot, Node.js, Express.js, React

Developer Tools: Git, Docker, Linux, CLI, PostgreSQL, RESTful APIs, CI/CD (GitHub Actions)

WORK EXPERIENCE

Software Engineer Intern — *VLI-EV* *Dec. 2025 – Present*

- Owned end-to-end AWS ingestion pipeline (**IoT, Firehose, S3, Athena**) handling ~**1.8M+** records / day
- Reduced cloud costs **78%** by optimizing service configuration, partitioning strategy, and data architecture
- Collaborated with hardware, leadership, and external partners to integrate and standardize BMS telemetry

Autonomous Research Assistant — *Real-Time Embedded & Controls Systems (RTEC) Lab* *Jan. 2025 – Present*

- Built **C++** localization systems for autonomous vehicles, improving lap times by **15%** over **100+** runs
- Replaced LiDAR-based **AMCL** with **Isaac ROS vSLAM**, reducing sensor cost while maintaining localization
- Integrated open-source object detection to classify dynamic obstacles in the **F1TENTH** platform

Instructional Student Assistant — *San Diego State University* *Aug. 2025 – Present*

- Graded **250+ Python and C/C++** submissions for **algorithms** and **operating systems** courses with feedback
 - Developed and maintained automated grading workflows and CLI testing scripts, reducing grading by **30%**
 - Supported **150+ students** through office hours with C/C++ debugging, OS concepts, and algorithms
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LEADERSHIP EXPERIENCE

Vice President — *App Development Club* *Aug. 2024 – Present*

- Led workshops on **Git, React**, Vue, Next.js, and **Spring**, guiding **20+** members in full-stack development
 - Presented on REST APIs, Spring Boot, and CLI, helping members better interview and project skills
 - Led **15+** building a non-profit farmers market website with auth, database, multi-role login, and payments
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PROJECTS

DevTrackr — *Task Management Backend Service* *FastAPI | PostgreSQL | Docker*

- Developed a **RESTful API** with **PostgreSQL** and SQLAlchemy ORM for secure, persistent task management
- Implemented **JWT** authentication and analytics for task summaries, category trends, and weekly insights
- Automated CI/CD with Docker Compose, pytest, and **GitHub Actions** for reliable deployment

The Crash App — *Innovate 4 SDSU Hackathon Winning Project* *Next.js | TypeScript | Tailwind | React.js*

- Built a **Next.js/TypeScript** app for hit-and-run reporting with image uploads and plate/timestamp data
- Designed structured backend storage with plate-lookup and verification workflows to validate reports

PrepPal — *AWS x Cal Poly DxHub Hackathon Winning Project* *AWS Bedrock | React | TypeScript*

- Built a full-stack AI interview coach using **Next.js** and **FastAPI**, with AWS **S3/DynamoDB/Bedrock**
- Implemented real-time audio/video processing pipelines using STT, TTS, and OpenCV for fast feedback

Send It or Shred It — *LPL Hackathon Project* *FastAPI | AWS Bedrock | RAG*

- Developed a **FastAPI** backend to analyze financial documents for SEC/FINRA compliance using **RAG**
- Implemented PDF, DOCX, and OCR pipelines with PyMuPDF and Tesseract for text extraction
- Integrated **AWS Bedrock** and **S3** for scalable inference, storage, and automated report generation